Luca Repupilli

Software Engineer

Macerata, Italy · Open to relocation
lucarepupilli@gmail.com · +39 331 3237292
GitHub · LinkedIn · Portfolio



SUMMARY

Computer Science graduate and aspiring AI engineer, now pursuing an MSc in Artificial Intelligence to specialize in AI & agentic systems. Hands-on experience building multi-agent LLM applications that combine tool-execution pipelines with Retrieval-Augmented Generation (RAG), and designing RESTful backends in Java / Spring Boot and C# / ASP.NET Core. Shipped a conversational multi-agent assistant (LangChain / LangGraph) as a bachelor's thesis, and delivered LLM automation workflows during an industry internship. Comfortable across Python, Java, and C#, and eager to grow as an AI engineer in a team that builds real-world, high-impact software.

TECHNICAL SKILLS

AI & Agentic: LangChain, LangGraph, multi-agent orchestration, Retrieval-Augmented Generation (RAG), prompt engineering, n8n
Languages: Python, Java, C#, C++, SQL
Backend & APIs: Spring Boot, ASP.NET Core, Entity Framework Core, JPA / Hibernate, REST API design, OpenAPI / Swagger, JWT authentication
Data & Persistence: relational databases (SQL), H2, ChromaDB, data modeling
Testing & Tooling: pytest, JUnit, Mockito, Pydantic, Docker, Gradle, Git / GitHub
Other: Streamlit, JavaFX, Arduino / embedded C++ (finite-state machines, interrupts), WordPress, Visual Paradigm

SELECTED PROJECTS

BdsLab Multi-Agent Assistant - Bachelor's Thesis

Python · LangChain · LangGraph · ChromaDB · Streamlit · Pydantic · pytest · Docker
github.com/bdslab/bdslab-rag-executive-agent - *Private repository, available on request.*

- Built a conversational multi-agent assistant that lets researchers drive six heterogeneous RNA-analysis command-line tools through plain natural language, removing the need to learn each tool's formats and options.
- Designed a supervisor-worker architecture in LangGraph that interprets a request, routes it to the right specialized agent, and chains tools into multi-step pipelines when a task depends on previous output.
- Implemented a RAG layer over a documentation knowledge base (ChromaDB + embeddings) to answer questions about the tools and the PhyloRNA taxonomy database.
- Made the system declaratively extensible (a new tool takes four localized steps, no changes to the graph or shared state), packaged it with Docker, and wrote a pytest suite with LLM calls mocked for deterministic tests.

AutomatedBlogPosting - LLM Content-Automation Workflow

n8n · LLM agents · Telegram Bot · FLUX (image generation) · WordPress
n8n workflow, demo available on request.

- Built an agentic workflow that turns a short Telegram chat into a publish-ready blog article: SEO-optimized, human-sounding copy plus a matching cover image generated with FLUX.
- Added a human-in-the-loop review over Telegram before publishing content as a WordPress draft. (Delivered during the KreaSistemi internship.)

HackHub2026 - Hackathon Management REST Backend

Java 21 · Spring Boot · JWT · OpenAPI/Swagger · JPA/Hibernate · JUnit · Mockito · Gradle · Visual Paradigm
github.com/SickLuca/HackHub2026

- Developed a RESTful backend covering the full hackathon lifecycle (events, teams, submissions, evaluations, reports, support requests) with a layered architecture (7 REST controllers), stateless JWT authentication, role-based access, and OpenAPI / Swagger documentation.
- Applied classic design patterns (Adapter, Builder, Strategy, Unit of Work) and covered business logic with JUnit / Mockito tests.
- Delivered as a team project: designed all UML diagrams (Visual Paradigm) and implemented the core modules and functionality, collaborating closely with teammates to keep the codebase consistent.

Library Catalog Web API

C# · ASP.NET Core · Entity Framework Core · SQL · Swagger · JWT
github.com/SickLuca/Unicam.Progetto.Libreria

- Built a Web API with JWT-secured operations for books, categories, and users, and filtered, paginated search (by category, title, author, publication date) over a layered, multi-project solution.

jbudget - Personal Budget Desktop App

Java · JavaFX · JPA / Hibernate · H2 · Gradle · Lombok
github.com/SickLuca/jbudget

- Built a desktop application for personal budget management with a JavaFX interface and a JPA / Hibernate + embedded H2 persistence layer.

ArduinoAlarmControlUnit - Embedded Alarm System

Arduino · C++ · Wokwi
github.com/SickLuca/ArduinoAlarmControlUnit

- Designed an alarm control unit using a finite-state-machine architecture and interrupt-driven programming with direct register manipulation for low-level hardware control (simulated on the Wokwi platform).

EXPERIENCE

Software Engineering Intern - KreaSistemi (*Corridonia, Italy*)

Oct 2025 - Dec 2025

- Developed and restyled client websites and e-commerce projects (WordPress / NicePage), deployed via Aruba Business.
- Built an AI agentic workflow in n8n: an LLM-powered Telegram bot that generates SEO-optimized articles and images and publishes them to clients' WordPress sites (see AutomatedBlogPosting).
- Developed LLM + RAG chatbots (n8n) for client websites that answer visitors' questions and assemble quotes to send by email.
- Strengthened client communication, problem-solving, and adaptability across multiple concurrent client projects.

Chef de Rang - Albergo Ristorante Apollo 17 (*Corridonia, Italy*)

Oct 2017 - Jan 2026

- Coordinated and led the dining-room staff while working full-time and later part-time throughout my studies.
- Developed strong teamwork, organization, and customer-facing communication skills under pressure.

EDUCATION

MSc in Computer Science - Artificial Intelligence (taught in English) - University of Camerino (UNICAM)

Oct 2026 · In progress

- Current courses: complex systems design, machine learning, multiagent systems lab, distributed calculus and coordination.

BSc in Computer Science - University of Camerino (UNICAM)

Oct 2021 · Graduated Jul 2026 · Grade: 102/110

- Core coursework: algorithms & data structures, software engineering (Java, Spring Boot), advanced programming paradigms (C# / ASP.NET Core), databases (SQL), operating systems, computer networks & security, languages & compilers, probability & statistics.

Scientific High School Diploma · Liceo Scientifico G. Galilei, Macerata

LANGUAGES

Italian: Native **English:** Professional working proficiency (CEFR self-assessed: Reading & Writing C1, Listening & Speaking B2)

SOFT SKILLS

Self-Confidence · Planning & Organization · Precision & Attention to Detail · Continuous Learning · Communication · Problem Solving · Teamwork · Leadership · Working Under Pressure · Customer Relations

ABOUT ME

After several years working in a busy restaurant, where I went on to lead the dining-room staff, I followed my passion for technology and enrolled in Computer Science at 23, working throughout my studies and falling in love with generative AI along the way. I'm driven by new challenges, and I believe the fastest growth comes from hands-on experience and the people around me. I'm deliberately critical of what I build, especially with AI: I want to understand what I ship all the way down, and I'll gladly redo something that doesn't meet my standards. My goal is to stand between the traditional developer and the prompt engineer, using AI with real engineering judgment. While I aim for AI engineering, I see value in every software role: what matters is to never stand still.

I hereby authorize the processing of my personal data contained in this CV pursuant to Legislative Decree 196/2003 and EU Regulation 2016/679 (GDPR).